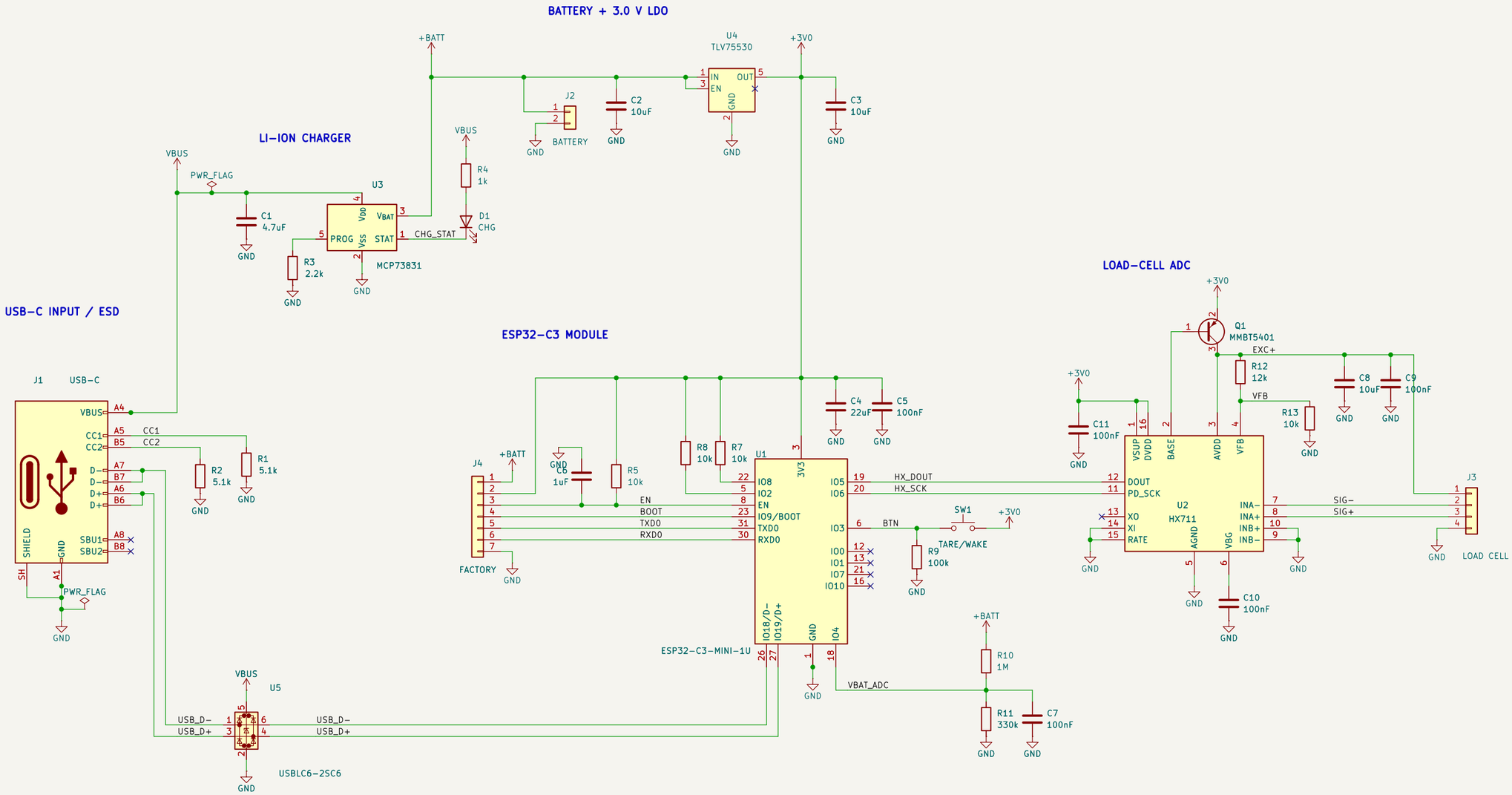


DESIGN NOTES
1. Battery-backed architecture: USB-C 5 V charges the 1S Li-ion through U3; the system always runs from VBAT via U4.
2. Charge current = 1000 V / R3 = 455 mA (0.15C for a 3000 mAh cell; stays inside the USB 2.0 500 mA budget).
3. U4 is a 25 uA-Iq 3.0 V LDO. 3.0 V keeps the ESP32-C3 in spec down to ~3.25 V cell voltage.
4. Load-cell excitation is regulated by Q1 and switched off when HX711 PD_SCK is held high (power-down).
   EXC+ = 1.25 V x (1 + R12/R13) = 2.75 V. RATE = 0 (10 SPS), internal oscillator.
5. VBAT_ADC = VBAT x 330k / 1.33M (4.2 V -> 1.04 V, ADC atten 2.5 dB). Divider draws 3.2 uA.
6. IO9 uses the module internal pull-up; pull it low on J4 while toggling EN for UART download.
7. Sleep budget: ESP32-C3 deep sleep ~5 uA + LDO 25 uA + divider 3 uA + HX711 PD <1 uA.

2-layer, 50 x 36 mm, JLCPCB PCBA
ESP32-C3-MINI-1U + HX711 load-cell ADC, USB-C Li-ion charging, 3.0 V low-Iq rail

Hector Jimenez

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Title: ESP32-C3 Smart Scale – Main Board

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